WELLSLING · DESIGN DECISION RECORD

WellSling concept selection

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DOCUMENT CONTROL

Document ID	WSL-DDR-003	Version	0.2
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsling		
Source	docs/decisions/0003-concept-selection.md @ 346fa2d		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Accepted: Amish chose option A and graft option (a) on 2026-10-25 ("i approve of your recommendations")
0.1	2026-10-25	Amish Chadha	Concept selection proposed (kit 1.9.1 upgrade); awaiting Amish

0003: Concept selection

- **Date:** 2026-10-25
- **Status:** Accepted. Amish Chadha (owner), 2026-10-25 12:43 CT: "i approve of your recommendations". Option A is carried forward (WSL-DEC-001 D15) and the adjusting chains stay (D16, option (a)); `concept_chosen: A`.

Context

Kit 1.8 requires at least two concept options with different working principles, scored against the user requirements (STANDARDS section 21). WellSling was designed before kit 1.8, so its design is option A and is not redesigned without Amish's go. Two further options were developed, modelled at detail level D3 and checked in the STEM check (WSL-STM-001): a portable gantry with two hand chain hoists (B) and sheerlegs with a draught haul (C). The scoring is in WSL-OPT-001, Table 2.

Options considered

Table 1. Options.

Option	Working principle	Weighted score (of 265)	Strengths	Weaknesses
A Tripod and hand winch	Own tripod and sheave; braked hand winch on one leg; two bands on an H-frame; slid-in bridge	224	Automatic brake; human power only; most margin; heaviest piece 32.0 kg	Setup time at its limit; one lift line, so the animal is levelled only by setting the chains before the lift
B Portable gantry and chain hoists	Box beam on two A-frames; one hand chain hoist per band	198	Chest and flank lifted separately; automatic hoist brakes	Beam assembled and lifted to 3.3 m; two operators pulling over the well mouth; 15 m of load chain per hoist; highest cost
C Sheerlegs and draught haul	Two leaning legs on a back guy; tractor hauls through blocks, with brake turns on a post	177	Fast lift; rope length cheap	No automatic brake; tractor jerk risk; guy anchor holding unknown; three anchors to set

Recommendation

Decided by Amish on 2026-10-25, as recommended: carry option A forward. B is 11.6 % behind, outside the 10 % band, so a single option is recommended rather than two. Graft B's best idea, separate adjustment of the chest and flank bands, by replacing one band's adjusting chains with a 0.75 t lever hoist if the TRL 4 dummy trial shows the animal hanging nose or tail down (WSL-DEC-001 D16, decided 2026-10-25: option (a), the adjusting chains stay and the dummy trial decides whether to try the lever hoist).

Consequences

- A is confirmed: no change to the design; `concept_chosen` in `project.yaml` is A.

- If B or C is chosen: the chosen option's model in `cad/src/model.py` becomes `CHOSEN`, and the drawings, priced BOM, build plan, calculation note and requirements statuses are redone for it.